1)

	Model	Prototype
D	0.2	0.3
ρ	1.23	1030
μ	1.79×10^{-5}	1.20×10^{-3}
V	?	1
F_D	25	?

$$F_D = f(D, V, \rho, \mu)$$

Expressing everything in basic dimensions:

$F_D = MLT^{-2}$

$\left.\begin{aligned} D &= L \\ V &= LT^{-1} \\ \rho &= ML^{-3} \end{aligned}\right\}$ repeating variables

$\mu = ML^{-1}T^{-1}$

Using ρ, V, D as repeating variables:

$$\begin{aligned} F_D \rho^a V^b D^c &= MLT^{-2} M^a L^{-3a} L^b T^{-b} L^c \\ &= M^a L^{-3a+b+c} T^{-b} MLT^{-2} \\ &= M^{a+1} L^{-3a+b+c+1} T^{-b-2} \end{aligned}$$

$$\begin{aligned} a &= -1 \\ -3a+b+c &= -1 \\ -b &= 2 \end{aligned} \Rightarrow$$

Solving:

$a = -1$

$b = -2$

$c = -2$

$$\therefore \pi_1 = \frac{F_D}{\rho V^2 D^2}$$

1) $\mu \rho^a V^b D^c = M^a L^{-3a+b+c} T^{-b} M L^{-1} T^{-1}$
$= M^{a+1} L^{-3a+b+c-1} T^{-b-1}$

$a = -1$
$-3a+b+c = 1$
$-b = 1$

$\Longrightarrow$ Solving:
$a = -1$
$b = -1$
$c = -1$

$\therefore \pi_2 = \dfrac{\mu}{\rho V D}$

$$\therefore \frac{F_D}{\rho V^2 D^2} = f\left(\frac{\mu}{\rho V D}\right)$$

a) $$\frac{\mu_m}{\rho_m V_m D_m} = \frac{\mu_p}{\rho_p V_p D_p}$$

$$\frac{1.79 \times 10^{-5}}{1.23 \times V_m \times 0.2} = \frac{1.20 \times 10^{-3}}{1030 \times 1 \times 0.3}$$

$$V_m = \frac{18437}{984}$$

$$\approx 18.7 \, ms^{-1}$$

b) $$\frac{(F_D)_m}{\rho_m V_m^2 D_m^2} = \frac{(F_D)_p}{\rho_p V_p^2 D_p^2}$$

$$\frac{25}{1.23(18.7)^2(0.2)^2} = \frac{(F_D)_p}{1030 \times 1^2 \times 0.3^2}$$

$$(F_D)_p = 134.1727514$$

$$\approx 134.17 \, N$$

2)

	Model	Prototype
V	?	65
F_D	400	?

From question 1,

$$\frac{F_D}{\rho V^2 D^2} = f\left(\frac{\mu}{\rho V D}\right)$$

$$\frac{\cancel{\mu_m}}{\cancel{\rho_m} V_m D_m} = \frac{\cancel{\mu_p}}{\cancel{\rho_p} V_p D_p}$$

$$\frac{1}{V_m D_m} = \frac{1}{65 D_p}$$

$$\frac{D_p}{D_m} = \frac{V_m}{65}$$

$$5 = \frac{V_m}{65}$$

$$V_m = 325 \text{ km h}^{-1}$$

2a) $$\frac{F_{D,m}}{\cancel{\rho_m} V_m^2 D_m^2} = \frac{F_{D,p}}{\cancel{\rho_p} V_p^2 D_p^2}$$

$$\frac{400}{325^2 D_m^2} = \frac{F_{D,p}}{65^2 D_p^2}$$

$$\frac{400}{325^2} = \frac{F_{D,p} D_m^2}{65^2 D_p^2}$$

$$\frac{400}{325^2} = \frac{F_{D,p}}{65^2}\left(\frac{1}{5}\right)^2$$

$$F_{D,p} = 400N$$

b) $P = Fv$

$$= 400\left(\frac{65\times10^3}{60^2}\right)$$

$$= 7222.2\dot{2}$$

$$\approx 7220 W$$

3a) $Fr = \frac{V}{\sqrt{gl}}$

	Model	Prototype
V	?	9.26

$$\frac{V_m}{\sqrt{gl_m}} = \frac{V_p}{\sqrt{gl_p}}$$

$$\frac{V_m^2}{\cancel{g}l_m} = \frac{V_p^2}{\cancel{g}l_p}$$

$$V_m^2 = V_p^2\left(\frac{l_m}{l_p}\right)$$

$$V_m = V_p\sqrt{\frac{l_m}{l_p}}$$

$$V_m = 9.26\sqrt{\frac{1}{50}}$$

$$= 1.309561759 \, ms^{-1}$$

$$\approx 1.31 \, ms^{-1}$$

3b) From question 1, replacing D with L

$$\frac{F_D}{\rho V^2 L^2} = f\left(\frac{\mu}{\rho V L}\right)$$

$$\frac{F_{D,m}}{\rho_m V_m^2 L_m^2} = \frac{F_{D,p}}{\rho_p V_p^2 L_p^2}$$

$$\frac{\cancel{\rho_p} V_p^2 L_p^2}{\cancel{\rho_m} V_m^2 L_m^2} = \frac{F_{D,p}}{F_{D,m}}$$

$$\frac{9.26^2}{1.31^2}\left(\frac{50}{1}\right)^2 = \frac{F_{D,p}}{F_{D,m}}$$

$$\frac{F_{D,p}}{F_{D,m}} = 125000$$

4)

$$Re = \frac{\rho U L}{\mu}$$

	Model	Prototype
P	1300	101
U	?	385

$$PV = mRT$$

$$\frac{m}{V} = \frac{P}{RT}$$

$$\rho = \frac{P}{RT}$$

$$Re_m = Re_p$$

$$\frac{\rho_m U_m L_m}{\cancel{\mu_m}} = \frac{\rho_p U_p L_p}{\cancel{\mu_p}}$$

$$\frac{P_m U_m L_m}{\cancel{RT_m}} = \frac{P_p U_p L_p}{\cancel{RT_p}}$$

$$\frac{U_m}{U_p}\frac{L_m}{L_p} = \frac{P_p}{P_m}$$

$$\frac{L_m}{L_p} = \frac{101}{1300}\left(\frac{U_p}{U_m}\right)$$

4) Range of U_m:

$$0.8U_p \leq U_m \leq 1.2U_p$$

$$\therefore \frac{101}{1300}\left(\frac{1}{1.2}\right) \leq \frac{L_m}{L_p} \leq \frac{101}{1300}\left(\frac{1}{0.8}\right)$$

$$\therefore 0.06474358974 \leq \frac{L_m}{L_p} \leq 0.09711538462$$

$$\therefore 0.0647 \leq \frac{L_m}{L_p} \leq 0.0971$$